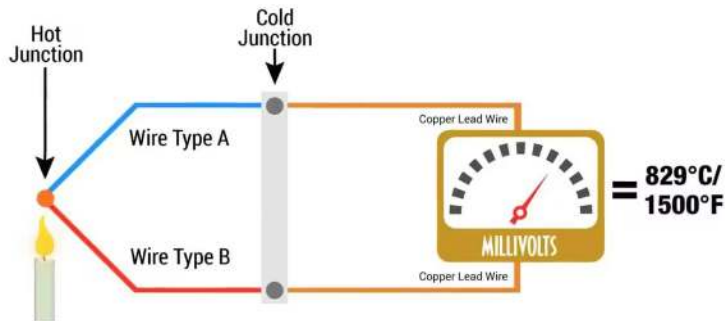


Introduction

The objective of the second laboratory on temperature sensors was to build upon the previous lab (which focused on thermistors and other basic temperature sensors) and to further explore thermocouples and temperature control systems. Class discussions and handouts were used as references throughout this experiment. Detailed information about each specific sensor tested is provided in the summaries later in the report. As discussed in class, there are many types of temperature measurement sensors. Industrially, the most common (and the ones focused on in this lab) are the thermistor, RTD (Resistance Temperature Detector), IC-based sensors, and thermocouples. The content of this lab report consists of summaries for each sensor system covered in the lab.

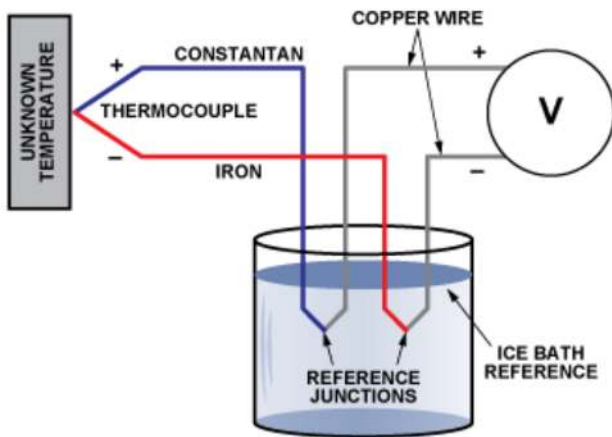
Part 1: Thermocouple summary

A thermocouple is a sensor for measuring temperature. Essentially, it is just two wires made of different metals joined together at one end (called the hot junction). When the joined end is heated or cooled, a tiny voltage is created between the two wires. The voltage changes depending on the temperature difference between the hot end (where you are measuring) and the other ends (where the wires connect to your meter). Thermocouples are great because they have a good temperature range and can be used for a lot of practical applications that other sensors cannot endure. Depending on what metal is used for the metal wires, a thermocouple is capable of measuring temperatures in the range -200C to +2500 C (<https://www.analog.com>). They are robust (immune to shock and vibrations, so they can be used in unstable and hazardous environments) (<https://www.analog.com>). They also do not add extra heat due to self heating, because they require no execution power, and they have a quick response to temperature changes. As far as disadvantages, they are prone to inaccuracies, susceptible to noise and need more maintenance, and are subject to some measurement hurdles.

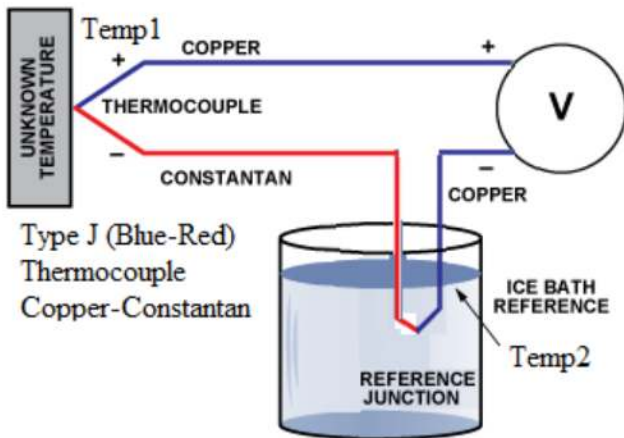


IMG1. Image of an example of a type T thermocouple, from <https://www.dwyeromega.com>. The metals joined together at one end (the hot junction) is the sensing tip. The other end (where the metal meets the copper) is called the reference or cold junction.

We can make our own thermocouple using either type T (copper + constantan, blue and red wires) or type K (chromel + alumel, yellow and red wires). Note that to follow U.S. standards for thermocouples, red is always the negative wire.

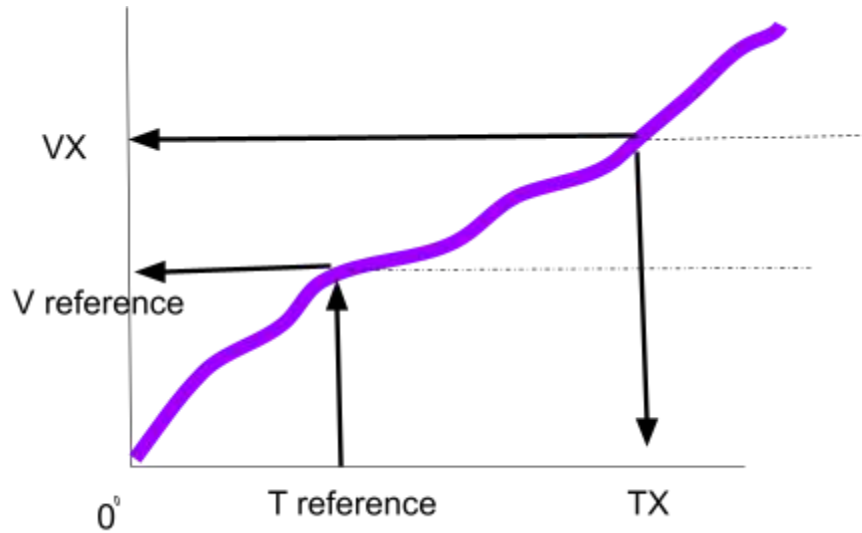


IMG 2. Note that there is no copper component. This setup is a bit complicated because the junctions must be held at a known temperature (ice bath at 0 degrees C) to eliminate their effect on the set up. This is why the set up in img 3 is used. This image is from Professor Herder's lab handout, using the basic thermocouple setup from <http://www.analog.com>.



IMG 3. This type T thermocouple is used to simplify the system. It is made up of copper and constantan, which eliminates one junction since the meter has copper leads allowing for a simpler set up. In the lab, temp 1 and temp 2 can be any temperature desired - but 0, 25 and 100 degrees C are nicer and lead to an easier set up. This image is from Professor Herder's lab handout, using the basic thermocouple setup from <http://www.analog.com>.

The wires are joined together (by twisting or soldering) and that joined point is the sensing tip. The voltage that appears on the other end depends on how hot or cold the tip is compared to your meter. We then can use tables (like NIST thermocouple tables) to convert that voltage into the temperature at the tip. This voltage produced depends on the temperatures at both the measurement junction and the reference junction. Since the thermocouple is a differential device as opposed to an absolute temperature measuring device, this means that the reference junction temperature must be known to get an accurate absolute temperature reading. That process of using a reference junction and compensating for it is called "reference junction compensation" (or cold junction compensation, shortened as CJC). CJC is needed because a thermocouple only measures the difference between the hot and cold junctions, so we need to correct for the actual cold junction temperature to get the true hot junction temperature.



IMG 4.

How does CJC work? The purple curve shows how voltage changes with temperature. Thermocouple produces a voltage: $V_{thermistor} = V_X - V_{reference}$, where V_X is the voltage that would exist if the cold junction was at $0^\circ C$, and $V_{reference}$ is the voltage caused by the actual cold junction temperature.

Using the type T set up, we took 4 voltage measurements



IMG 5.



IMG 6.

Starting Condition:

A Type T thermocouple was connected to the Fieldpiece SC260 meter using a yellow thermocouple plug with the meter set to the °F/°C temperature mode. The display initially showed 85°F (approximately 29°C), which corresponds to room temperature. Two measurements of room temperature were taken. The slight change in the second image might have occurred because the thermocouple junction was warmed slightly by contact with a finger. At this stage, however, the thermocouple is simply measuring the temperature of the air/room.



IMG 7.

The measured temperature was 35.8°F, which is close to the expected/ ideal ice bath temperature of 32°F (0°C).



IMG 8.



IMG 9.

These values can be tabulated and checked with the NIST tables at https://srdata.nist.gov/its90/download/type_j.tab.

Part 2:RTD testing

For the RTD testing part of the lab, we worked with a Resistance Temperature Detector (RTD), which is typically made from platinum (though other metals like nickel or copper can also be used). Since an RTD's resistance changes with temperature, we could measure temperature in two ways: by directly measuring its resistance or by setting up a voltage divider circuit and measuring the voltage across the RTD. The second method required a stable and accurate voltage source. In the lab, we measured the RTD's resistance both in air and when holding it between our fingers to observe how the resistance increased with temperature. Because platinum RTDs are usually $100\ \Omega$ at $0\ ^\circ\text{C}$, we used a low test voltage (around 2 V) and a second resistor of about $100\ \Omega$ in series to avoid self-heating.

When we hold the wire, it is 87 degrees, while when we let go of the wire it is 91.

https://youtube.com/shorts/_5TgREN-UVs



IMG 10. *At room temperature*

We also reviewed the 3-wire RTD circuit, which helps cancel out lead resistance, and compared it to the more precise 4-wire resistance measurement setup commonly used in high-accuracy systems.

Part 3: Peltier TEC summary

One of the temperature demonstrations in the lab involved the Peltier Thermoelectric Cooler (TEC). This device is a special application of the thermocouple that actually operates in the reverse manner of a traditional thermocouple. Instead of generating a voltage when heat is applied, the TEC uses an electrical voltage and current to create a temperature difference between its two sides. This means that as a result, one side becomes cold while the other becomes hot, allowing the system to cool objects below room temperature.



IMG 11.

In the first link below, we can see the temperature decreasing as voltage and current are applied to the Peltier Thermoelectric Cooler (TEC). This demonstrates that the cold side of the TEC is actively dropping in temperature, which confirms that electrical energy is being converted into a temperature difference. As the temperature reading falls, it indicates the surface is getting colder. That is a direct example of the Peltier effect and the reverse of how a normal thermocouple operates. The second link shows the opposite side, where the temperature increases as heat is released. This happens because when one side absorbs heat the other side must dissipate that heat, thus becoming hotter. Together both videos visually demonstrate the dual heating and cooling behavior of the TEC, showing how it transfers heat from one side to the other side.

Temperature decreasing

<https://youtube.com/shorts/jsh1TDV5cYc?feature=share>

Temperature increasing

https://youtube.com/shorts/_5TgREN-UVs?feature=share

Industrial Temperature controllers: summary and parts A-C

Industrial temperature controllers are important tools in many measurement and control systems. In industry we often don't just want to measure temperature — we also want to control it. These controllers automatically turn heating or cooling systems on or off to keep a temperature within a desired range. Temperature controllers are one of the most common types of controllers used in industrial applications, and this section of the lab demonstrated to us about how temperature measurement ties into control systems, as we aren't just measuring but seeing how the controller can respond to that measurement to regulate temperature automatically. Industrial temperature controllers come in two basic forms. One form is the simplistic and inexpensive STC-1000.



IMG 12. STC-1000. More information can be found at

<https://images-na.ssl-images-amazon.com>



IMG 13. *Image from part 1- looking at the STC-1000, determining the connection information, and powering it up to see how it works.*

STC1000 Operation Instruction

Main function

Switch the modes between cool and heat; Control temperature by setting the temperature set value and the difference value; Temperature calibration; Refrigerating control output delay protection; Alarm when temperature exceeds temperature limit or when sensor error.

Specification and size

- ◆ Front panel size: 75(L)x34.5(W)(mm)
- ◆ Mounting size: 71(L)x29(W)(mm)
- ◆ Product size: 75(L)x34.5(W)x85(D)(mm)
- ◆ Sensor length: 1 m(include the probe)

Technical parameters

- ◆ Temperature measuring range: -60℃~120℃
- ◆ Resolution: 0.1℃
- ◆ Accuracy: ±1℃(-50℃~70℃)
- ◆ Power supply: 12V 24V 110-220VAC 50Hz/60Hz
- ◆ Sensor: NTC sensor(1PC)
- ◆ Power consumption: <3W
- ◆ Relay contact capacity: Cool(10A/220VAC); Heat(10A/220VAC)
- ◆ Ambient temperature: 0℃~60℃
- ◆ Relative humidity: 20~85%(No condensate)
- ◆ Storage temperature: -30℃~75℃

panel instruction



Display instruction: Three-digit LED + Minus digit + Status indicator light (Status indicator light (Cool, Heat) + set indicator light (Set))

Key instruction: "S" key: the key to set; "▲" key: Up key; "▼" key: Down key; "⏻" key: the key to turn on and off the power.

Indicator light status instruction

Indicator light	Function	Note
Cool indicator	On: Refrigeration starts; Off: Refrigeration stops; Flash:	Cool. Heat indicator light can not be "on" status simultaneously
Heat indicator	On: heating starts; Off: heating stops	
Set indicator li	On: parameter setting status	

Key operation instruction

1. The way to check parameter:

Under normal working status, press "▲" key it displays temperature setting value; press "▼" key it displays the difference value.

2. The way to set parameter:

Under controller normal working status, press "S" key for 3s or more to enter parameter modifying mode, and the "Set" indicator light on, screen displays the first menu code "F1".

Press "▲" key or "▼" key to adjust up and down and display the menu item and the code of the menu item. Press "S" key to display the parameter value of the current menu. Press both "S" key and hold "▲" key or "▼" key simultaneously to choose and adjust the parameter value of the current menu value promptly. After finishing the setting, press and release the "⏻" key instantly to save the parameter modified value and return to display the normal temperature value. If no key operation within 10 seconds, system won't save modified parameter, screen back to display normal temperature.

3. Restore system data

When electrified, system will check itself, screen will display "Er" if error exit, please press any key at this time, and it restores default value and enter into normal working mode, it is advised to reset the parameter value under such conditions.

IMG 14. Operation instructions

Operation instruction

Under controller normal working status, press and hold "⏻" key for 3s can turn off the controller; Under controller "off" status, press and hold "⏻" key for 1s can turn on the controller.

Under the controller normal working status, screen displays the current measuring temperature value; also the controller can also switch the working mode between heating and cooling.

Controller starts refrigerating with cool indicator light on when the measuring temperature value > temperature set value + difference value, and the refrigerating relay is connected; If the "Cool" indicator light flashes, it indicates the refrigerating equipment is under compressor delay protect status; when the measuring temperature values temperature set value, the Cool indicator light on, and refrigerating relay disconnects.

System starts heating when the measuring temperature value < the temperature set value - difference value, and the "Heat" indicator light on, the heat relay connects; When the measuring temperature 2 temperature set value, the "Heat" indicator light is off, and the heat relay disconnects.

Menu instruction

Code	Function	Set range	Default	Note
F1	Temperature set value	-60.0°C~120.0°C	10°C	
F2	Difference set value	0.1~30.0°C	3°C	
F3	Compressor delay time	1~10 minutes	3	
F4	Temperature calibration value	-10.0°C~10.0°C	0°C	

Error description

Alarm when sensor error: Controller activate the sensor error alarm mode when sensor open circuit or short circuit, all the running status is closed off with the buzzer alarms, and the nixie tube displays "EE", press any key can cancel alarm sound, system back to display the normal temperature when the error and the fault is cleared.

Alarm when the measuring temperature exceeds temperature measuring range: Controller activates the error alarm function when the measuring temperature exceeds the temperature measuring range, all the running status is closed off with the buzzer alarms, and the nixie tube displays "HH", Press any key can cancel alarm sound, system back to display the normal working mode when the temperature restore to normal measuring range.

Safety Regulations

★ Danger:

1. Strictly distinguish the sensor down-lead, power wire and output relay interface from one another, and prohibit wrong connections or overloading the relay.
2. Dangers: Prohibit connecting the wire terminals without electricity cut-off.

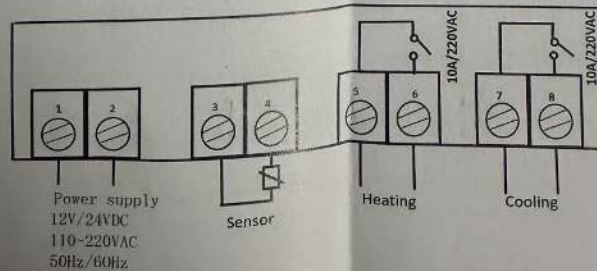
★ Warning:

Prohibit using the machine under the environment of over damp, high temp., strong electromagnetism interference or strong corrosion.

★ Notice:

1. The power supply should conform to the voltage value indicated in the instruction.
2. To avoid the interference, the sensor down-lead and power wire should be kept a proper distance.

Wiring diagram



IMG 15. Operation instructions

The STC-1000 is a temperature controller, which is basically a low cost digital thermostat used for automatic heating or cooling control. It is a small digital thermostat that can automatically switch between heating and cooling modes. We set a target temperature (set point) and a difference value (hysteresis, the range around the set point that triggers on/off actions). It can turn on a heater when the temperature drops too low, turn on a fan when the temperature gets too high or even give off an alarm when the temperature exceeds a limit or when a sensor error occurs. Operations instructions, wiring diagram, safety notes, error descriptions, and specifications of the device can be shown in images 14 and 15. The connection information was determined, and then it was powered up to see how it works. Next, we picked one of the controllers listed below and used the manual to first answer the following questions (part a), then connect a sensor to the unit (part b), configure the setup documenting the settings (push button type procedure) to display a temperature on the unit (part c), and note power and a sensor will have to be added of course to do this.

Part A

A. List any special terms or abbreviations you had never seen or heard of before reviewing a controller datasheet or manual.

While reviewing the STC-1000 manual (images 14 and 15), we saw new terms such as PV (Process Value), SP (Setpoint), F1–F4 parameters, and hysteresis (difference value). These define how the controller senses and regulates temperature.

1. What kind of Power Supply is required to operate the unit?

AC 110–220 V

2. What kind of output does the unit have, options are often "digital" relay or switched transistor or even continuous "analog" control of a voltage or current

The STC-1000 has two relay outputs—one for heating and one for cooling.

3. If the unit has digital control does it have pulsed or PWM type control or simple ON-OFF control

The controller uses simple ON/OFF control with a pulsed delay for compressors (parameter F3).

4. If the unit has analog control does it have PID control?

The STC-1000 does not have PID or analog output; control is binary (ON/OFF).

5. What types of sensors can be used with the units, list all types

It uses an NTC thermistor temperature probe (10 k Ω @ 25 °C).

6. If the unit mounts in a control panel, what DIN size is it, what does the DIN number mean?

The unit is a mini panel mount type, about 1/16 DIN (48 × 48 mm); DIN refers to standardized mounting sizes for control panels.

7. Is the SP or Setpoint Variable of the controller set digitally or by adjustable element such as a Potentiometer?

The SP is adjusted digitally using the S button and arrow keys.

8. Is the PV or Process Variable of the controller displayed digitally?

The measured temperature is displayed digitally on the LED screen in °C.

9. Does the unit have any Automatic features such as "auto-tuning"?

It includes auto compressor delay (F3) and calibration offset (F4) but no auto-tuning.

10. Are there any protective limits for temperature or output V or I to protect the unit or system being controlled, list them if there are.

Parameters F1–F4 include built-in temperature and delay limits to prevent damage to heaters, coolers, and compressors.

11. From the manual document a basic connection or schematic diagram of the usage for the controller, it is ok to scan or screen capture a diagram from a pdf from the manual. Numbers: 2, 3, and 8 are particularly capable and modern units

See image 15 for this information

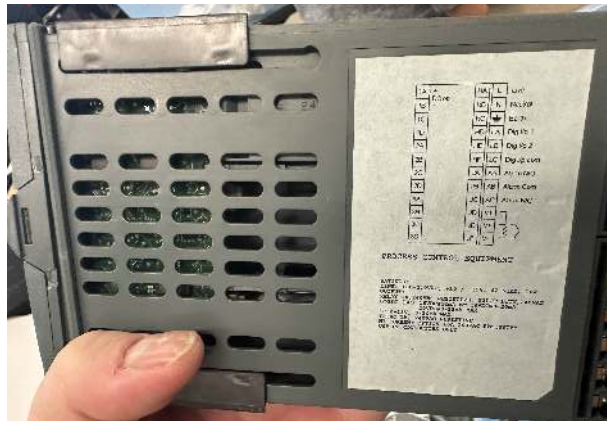
Part B & C

A sensor was connected to the unit, and the set up was configured to display a temperature on a unit (power and a sensor is added to do this).

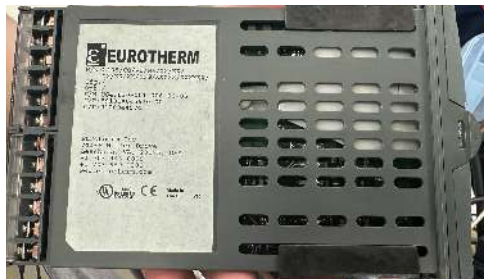
An NTC sensor probe was connected to the STC-1000 through terminals 3 and 4. The controller was then powered through terminals 1 and 2 (AC 110 V). After powering up the display immediately showed the current room temperature (~28.9 °C), confirming that the sensor was wired and functioning correctly.



IMG 16.



IMG 17.



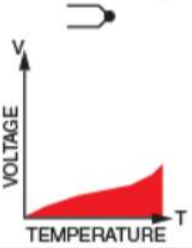
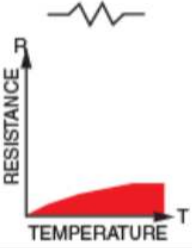
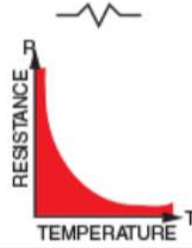
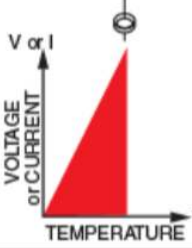
IMG 18.

These three photos show the Eurotherm 2408 industrial temperature controller, which is part of the industrial temperature controller section of the lab. The Eurotherm is a device used in factories and labs to automatically monitor and control temperature using sensors such as thermocouples or RTDs. The first photo shows the front display and

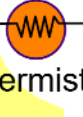
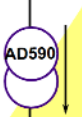
buttons, which let you view the current temperature and set a target temperature (setpoint). The second photo shows the back side with terminal connections, where power, sensors, and outputs are wired into the controller. The third photo shows the wiring diagram label. That provides a clear guide for connecting the correct wires to their terminals for power input, sensor input, and control outputs. Together, these photos represent how the Eurotherm is used to measure and regulate temperature in a system by reading from a sensor and activating heating or cooling components as needed.

Conclusion

When selecting between which to use (thermocouples rtds thermistors etc) consider the characteristics and costs of the available sensors. Thermocouples can general measure temperatures over wide temperature ranges, are inexpensive, and are rugged, but not as accurate or stable as RTDs and thermistors. As far as RTDs go, they are stable and have a fairly wide temperature range, but are not as rugged and inexpensive as thermocouples. Since they require the use of electric current to make measurements, RTDs are actually vulnerable to the inaccuracies that come with some added self heating. They also require 4 wire measurement systems to compensate for extra added resistance in the system. Because of that, thermistors tend to be a bit more accurate then RTDs and thermocouplers, but thermistors have a more limited temperature range and are also still subject to added extra heat.

	Thermocouple 	RTD 	Thermistor 	I. C. Sensor 
Advantages	- Self-powered - Simple - Rugged - Inexpensive - Wide variety - Wide temperature range	- Most stable - Most accurate - More linear than thermocouple	- High output - Fast - Two-wire ohms measurement	- Most linear - Highest output - Inexpensive
Disadvantages	- Non-linear - Low voltage - Reference required - Least stable - Least sensitive	- Expensive - Current source required - Small ΔR - Low absolute resistance - Self-heating	- Non-linear - Limited temperature range - Fragile - Current source required - Self-heating	$T < 200^\circ\text{C}$ - Power supply required - Slow - Self-heating - Limited configurations

IMG 19. The table is from www.omega.com

Agilent Technologies Classroom Series		
Summary: Absolute T devices		
 RTD *Most accurate *Most stable *Fairly linear *Expensive *Slow *Needs I source *Self-heating *4-wire meas.	 Thermistor *High output *Fast *2-wire meas. *Very nonlinear *Limited range *Needs I source *Self-heating *Fragile	 I.C. AD590 LM34/5 *High output *Most linear *Inexpensive *V or I source *Limited range *Needs V source *Self-heating *Previously Limited Variety

IMG 20. From Prof. Gerald Herder, shared in power point in his lecture for 4201, Keysight/Agilent App Note "Practical Temperature Measurements"

Below is a list of real world industrial temperature control units made by different manufacturers.

1. Omron EC5K-AA1-500

<http://industrial.omron.com.br/uploads/arquivos/M14H083E102A.pdf>

<http://mikrokontrol.rs/proizvodi/pdf/H078-E1-03D+E5CK+UsersManual.pdf>

2. Moore Industries 535 <http://www.mmcontrol.com/Moore/powers-535.php>

http://www.miinet.com/Portals/0/Manuals/535_Installation_Manual_Moore_Industries.pdf

3. Moore Industries 330

http://www.miinet.com/Portals/0/Manuals/330R_Installation_Manual_Moore_Industries.pdf

4. Watlow Series

965 <http://www.watlow.com/en/Resources-And-Support/Technical-Library/User-Manuals>

5. Watlow 988A Temperature Controller

<https://www.watlow.com/-/media/documents/user-manuals/988ev.ashx>

6. APC TEMP/HUMIDITY SENSOR MODEL NUMBER:AP9520TH

http://www.apc.com/salestools/ASTE-6Z5QC9/ASTE-6Z5QC9_R0_EN.pdf

7. Omega CN1A

<http://www.omega.com/pptst/CN1A.html>

<https://www.omega.com/manuals/manualpdf/M2796.pdf>

8. Omega CN132

<http://www.omega.com/pptst/CN132.html>

9. Omega PT Demo which has the CN8DP controller

<http://www.omega.com/pptst/CNPT-DEMO.html>

http://www.omega.com/pptst/CNPT_SERIES.html

10. Calcontrols 9500P

<http://www.mmcontrol.com/calcontrols/pdfs/CAL-9500P-Full-Manual-English-.pdf>

11. TEC or Thermal Electric Cooler type controller, very common in laser diode applications to

control the temperature to a constant value. WAVELENGTH TEMPERATURE CONTROLLER

https://www.allectronics.com/mas_assets/media/allectronics2018/spec/TRC-1500.pdf

12. EUROTHERM 2408

https://www.eurotherm.com/index.php?route=module/downloads/get&download_id=104

13. NEWPORT ELECTRONICS QUANTA MODELNUMBER: Q9020-MDF1

http://www.newportus.com/Pdf/Q2000O_9.pdf

14. NEWPORT IDP INFINITY D PROCESS METER

<https://www.newportus.com/Pdf/IDP.pdf>

<https://www.omega.com/manuals/manualpdf/M1676.pdf>

15. OMEGA MODEL DP25-S

<https://www.omega.com/manuals/manualpdf/M3598.pdf>

16. OMEGA MODEL DP41-S-S2 OR DP41-S-A

<https://www.omega.com/manuals/manualpdf/M1291.pdf>

17. OMEGA MODEL DP41-E-S2

<https://www.omega.com/manuals/manualpdf/M1297.pdf>14.

NEWPORT IDP INFINITY D PROCESS METER

<https://www.newportus.com/Pdf/IDP.pdf>

<https://www.omega.com/manuals/manualpdf/M1676.pdf>

18. OMEGA MODEL DP25-S <https://www.omega.com/manuals/manualpdf/M3598.pdf>

19. OMEGA MODEL DP41-S-S2 OR DP41-S-A

<https://www.omega.com/manuals/manualpdf/M1291.pdf>

Works cited:

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- STC-1000 Operating Instructions. (n.d.). *71VSFdFfszL.pdf*. Retrieved from
<https://images-na.ssl-images-amazon.com/images/I/71VSFdFfszL.pdf>